

Biliary tract cancer

Gallbladder · intrahepatic, perihilar and distal cholangiocarcinoma · **one systemic backbone, four different operations, and a molecular profile that depends entirely on where the tumor sits**

Compiled from the *Stanford Oncology Handbook 2025, Gallbladder Cancer and Cholangiocarcinoma* (Northness A, Heestand G) · NCCN · primary trial reports and FDA actions verified against PubMed, September 2026.

FOUR DISEASES IN ONE NAME

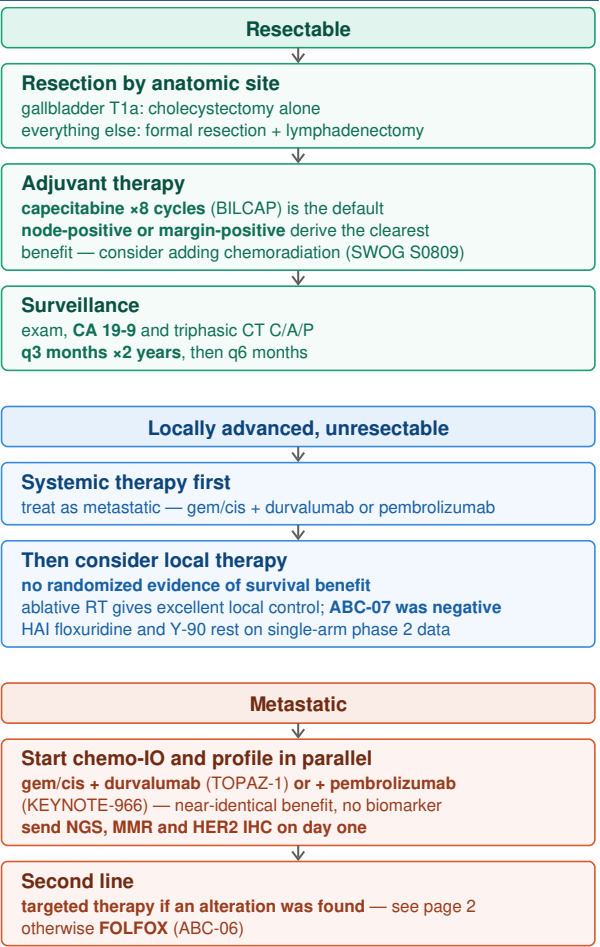
Site	Notes
Intrahepatic CCA ~10% of CCA	Arises proximal to the second-order ducts. The molecularly rich one — FGFR2 and IDH1 live here
Perihilar (Klatskin) ~40% of CCA	At the confluence of the right and left hepatic ducts. Resection planned by Bismuth-Corlette class
Distal CCA ~50% of CCA	Common bile duct up to, but not including, the ampulla
Gallbladder cancer	<5,000 cases/yr US; fundus 60% > body 30% > neck 10%. 1–2% are found incidentally at cholecystectomy for stones

~8,000 cholangiocarcinomas/yr in the US, median age ~70. **The subtypes share systemic therapy but differ in staging system, operation, and molecular profile** — which is why anatomic site has to be stated before anything else. **Risk:** CCA — PSC, HBV/HCV, liver flukes, male sex, obesity, diabetes, alcohol. **GBC** — cholelithiasis, gallbladder calcification, female sex, chronic Salmonella or H. pylori infection.

WORKUP

- **Labs:** CBC with differential, CMP, PT/INR, **CA 19-9**, HBV and HCV serologies
- **Imaging:** triphasic CT abdomen/pelvis or **liver-protocol MRI** + CT chest to complete staging. **PET/CT** to exclude occult disease when the plan is resection
- **Gallbladder:** start with ultrasound — go to triphasic CT or MRI for a **polyp >10 mm** or wall thickening not explained by cholecystitis
- **MRCP** to plan biopsy and stenting. **ERCP** to stent, decompress and brush — **brushings are often non-diagnostic, so go to EUS-FNA** when they are
- **Genomically profile every tumor**, and send **MMR, HER2 IHC and PD-L1**. This is not optional in biliary cancer — roughly a third of patients have a level 1/2 actionable alteration
- **If perihilar CCA might go to transplant, do NOT perform a transperitoneal biopsy** — tumor seeding disqualifies the patient from the protocol

TREATMENT PATHWAY



STAGING — THREE SEPARATE AJCC SYSTEMS

Site	T staging hinges on
Intrahepatic	Size and vascular invasion: T1a ≤5 cm / T1b >5 cm, both without vascular invasion · T2 vascular invasion <i>or</i> multiple tumors · T3 perforates visceral peritoneum · T4 direct extrahepatic invasion. N is simply N0 / N1
Perihilar	Depth then vessels: T1 confined to the duct · T2a into surrounding fat, T2b into hepatic parenchyma · T3 unilateral portal vein or hepatic artery branch · T4 main portal vein bilaterally, common hepatic artery, or second-order radicals with contralateral vessel
Distal	Millimetres of duct-wall invasion: T1 <5 mm · T2 5–12 mm · T3 >12 mm · T4 celiac axis, SMA or common hepatic artery
Gallbladder	Which side of the wall: T1a lamina propria, T1b muscularis · T2a peritoneal side, T2b hepatic side — the distinction drives the operation · T3 serosa, liver or one adjacent organ · T4 main portal vein / hepatic artery or ≥2 organs

Notes: perihilar, distal and gallbladder all use **N1 = 1–3 nodes, N2 = ≥4**. Intrahepatic uses N0/N1 only.

SURGERY — THE ONLY CURATIVE OPTION

- Most patients are unresectable at presentation, and **most resected patients still recur**
- **Resectability requires:** no retropancreatic or paraceliac nodes, no distant liver metastases, no main portal vein or hepatic artery invasion, no adjacent-organ invasion, no dissemination
- **Intrahepatic** → hepatic lobectomy; 5-yr OS 11–40%. **Perihilar** → en bloc resection by Bismuth-Corlette class. **Distal** → Whipple + extrahepatic duct; 5-yr OS 10–40%
- **Gallbladder: T1a is cured by simple cholecystectomy alone. T1b and above need en bloc cholecystectomy + resection of liver segments 4a/4b + regional lymphadenectomy**
- **Incidentally discovered gallbladder cancer beyond T1a must go back to the operating room** for the 4a/4b bed resection and lymphadenectomy
- **Liver transplant** is an option for highly selected **perihilar** CCA (Mayo protocol: mass ≤3 cm, no transperitoneal biopsy, no metastases) — intention-to-treat 5-yr survival ~53%

Biliary tract cancer

Page 2 — molecular profiling, which is where biliary cancer is won or lost

ALTERATION BY SITE — WHY SITE MATTERS

	ICCA	eCCA	GBC
FGFR2 fusion / rearrangement	~12%	~1%	~1–2%
IDH1	~20%	~1%	~1%
IDH2	~6%	0%	~0.5%
HER2 / ERBB2	~4%	~9%	~11–14%
BRAF V600E	~3%	~0%	~0%
KRAS, any	~12%	~38%	~10%
MSI-H / dMMR	~1.5%	~4%	~2%
BAP1	~19%	~2%	~1%
TP53	~21%	~44%	~60%

FGFR2 and IDH1 are intrahepatic diseases and are mutually exclusive. HER2 is a gallbladder and extrahepatic disease. Roughly a third of all biliary cancers carry a level 1/2 actionable alteration — but that is ~40% of intrahepatic versus ~15% of extrahepatic. Western cohorts (AACR GENIE); in East Asian series IDH1 and FGFR2 fall and KRAS rises.

LIVING WITH AN FGFR INHIBITOR

- **Hyperphosphatemia is on-target**, not idiosyncratic — FGFR1 blockade removes the FGF23 phosphaturic signal. It is a marker that the drug is working. **Pemigatinib ~60%, futibatinib ~85%, lirafugratinib ~21%**. Manage with a low-phosphate diet, **sevelamer**, and dose holds
- **Ocular toxicity**: serous retinopathy / retinal pigment epithelial detachment, plus dry eye in over 20%. **Baseline and serial ophthalmology with OCT** — roughly q2 months for 6 months, then q3 months
- **Nail and hair**: onycholysis in ~47%, alopecia in a third. Stomatitis, arthralgia and diarrhea are common
- **Resistance arrives at ~6–9 months** as **polyclonal secondary FGFR2 kinase-domain mutations**, classically the gatekeeper **V564F**. The covalent agents (futibatinib, lirafugratinib) retain activity against some but not all of these

DRIVER → THERAPY

Alteration	Therapy
FGFR2 fusion or rearrangement ~12% of ICCA	All second-line and beyond. Lirafugratinib (FGFR2-selective; approved 23 Sep 2026) — ORR 46%, DOR 11.8 mo, and notably less hyperphosphatemia (~21%) . Futibatinib ORR 42% (accelerated; label is intrahepatic only). Pemigatinib ORR 37% (accelerated, any site). Only fusions and rearrangements qualify — point mutations do not . FIGHT-302 moved pemigatinib to 1L on PFS but showed no OS gain , and the 1L indication is not approved. Infigratinib was withdrawn in 2024
IDH1 ~20% of ICCA	Ivosidenib after 1–2 prior lines (regular approval). PFS 2.7 vs 1.4 months, HR 0.37 — statistically emphatic, clinically modest. Unadjusted OS was not significant (10.3 vs 7.5 mo) because 70% crossed over ; crossover-adjusted HR 0.49
HER2 <i>GBC ~11–14%, eCCA ~9%</i>	Zanidatamab — the only BTC-specific approval (accelerated, Nov 2024), IHC 3+ only , 2L+: ORR 52% in IHC 3+, DOR 14.9 mo. T-DXd is covered by the tumor-agnostic IHC 3+ approval, but ILD reached 17% in the biliary cohort — higher than in most tumor types. Trastuzumab + pertuzumab is NCCN-listed but not FDA-approved here
BRAF V600E ~3% of ICCA	Dabrafenib + trametinib — tumor-agnostic accelerated approval. ROAR biliary cohort ORR 47–53%, PFS 9.0 mo
MSI-H / dMMR ~1–3%	Pembrolizumab (tumor-agnostic, now regular approval) — KEYNOTE-158 cholangiocarcinoma cohort ORR 41% , median OS 24.3 months. Dostarlimab is an alternative
TMB ≥10 mut/Mb ~4%	Pembrolizumab, tumor-agnostic accelerated approval
NTRK or RET fusion <1% each	Larotrectinib / entrectinib / repotrectinib for NTRK; selpercatinib for RET (tumor-agnostic, converted to traditional approval in 2026). Pralsetinib has no tumor-agnostic approval
KRAS G12C ~1%	No approved option in biliary cancer . Adagrasib showed activity in a small pan-tumor cohort (5/12); trial or off-label only

Biliary tract cancer

Page 3 — regimen dosing, the evidence behind adjuvant and local therapy, and trials

REGIMEN DOSING

Regimen	Dosing
Gem/cis + durvalumab <i>TOPAZ-1, 1L</i>	Gemcitabine 1000 mg/m² + cisplatin 25 mg/m² on days 1 and 8 , with durvalumab 1500 mg day 1 — q21d . Chemotherapy for up to 8 cycles, durvalumab continues
Gem/cis + pembrolizumab <i>KEYNOTE-966, 1L</i>	Same gem/cis backbone with pembrolizumab 200 mg day 1, q21d
FOLFOX <i>ABC-06, 2L</i>	Oxaliplatin 85 mg/m² day 1 + 5-FU 1200 mg/m²/day ×48 h continuous infusion from day 1 — q14d
Gem + nab-paclitaxel	Nab-paclitaxel 125 mg/m² then gemcitabine 1000 mg/m² , days 1, 8, 15 of a 28-day cycle . Single-arm phase 2 that missed its primary endpoint — an alternative, not a standard
Adjuvant capecitabine <i>BILCAP</i>	1250 mg/m² PO twice daily, days 1–14 of a 21-day cycle, ×8 cycles (6 months). Grade 3 hand-foot in ~20%
Adjuvant chemoRT <i>SWOG S0809</i>	Gemcitabine + capecitabine ×4 cycles, then capecitabine with radiation ×6 weeks (45 Gy nodal, 54–59.4 Gy tumor bed)
Ivosidenib	500 mg PO daily — IDH1-mutant, after 1–2 prior lines
Lirafugratinib	70 mg PO daily — FGFR2 fusion/rearrangement, previously treated, FGFR-inhibitor-naïve

READING THE ADJUVANT EVIDENCE HONESTLY

- **BILCAP is the standard of care but was a formally negative trial.** ITT median OS 51.1 vs 36.4 months, **HR 0.81, p=0.097** — **the primary endpoint was not met**. It became practice on the per-protocol analysis (HR 0.75) and a prespecified sensitivity analysis (HR 0.71). Worth knowing when someone asks why capecitabine
- **SWOG S0809 is single-arm** — 2-year OS 65% in resected extrahepatic CCA or gallbladder cancer with pT2-4, node-positive or margin-positive disease. Benchmark data, not a randomized comparison
- **Negative adjuvant trials:** PRODIGE-12 (GEMOX) and BCAT (gemcitabine) both failed. **ASCOT** showed adjuvant **S-1** improves 3-year OS (77.1% vs 67.6%) in an all-Japanese population — but **S-1 is not available in the US**
- **ACTICCA-1** (gem/cis vs capecitabine) has closed accrual and **has not yet read out** — it is the trial that may finally settle this

LIVER-DIRECTED THERAPY — WHAT IS ACTUALLY SUPPORTED

- **Nothing randomized.** Every option below rests on single-arm phase 2 or retrospective data
- **HAI floxuridine** via implanted pump (partial response 58%, median OS 25 months in a single-institution phase 2) — requires an experienced hepatobiliary program
- **Y-90 / TARE** concurrent with first-line chemotherapy: ORR 39%, median OS 22 months, 22% downstaged to surgery — but grade 3–4 toxicity was 71%, and **the one randomized trial, SIRCCA, was terminated**
- **Ablative radiation** gives excellent local control, but **ABC-07 randomized SBRT after chemotherapy and missed its PFS endpoint** (8.6 vs 9.0 months). The frequently quoted dose-escalation survival figures come from a retrospective series with heavy selection bias

KEY TRIALS

ABC-02	Gemcitabine + cisplatin vs gemcitabine — OS 11.7 vs 8.1 mo , HR 0.64. The backbone, since 2010
TOPAZ-1	Add durvalumab to gem/cis — OS 12.8 vs 11.5 mo, HR 0.80; 24-mo OS 24.9% vs 10.4%
KEYNOTE-966	Add pembrolizumab to gem/cis — OS 12.7 vs 10.9 mo, HR 0.83. Effectively interchangeable with TOPAZ-1
ABC-06	FOLFOX after gem/cis — OS 6.2 vs 5.3 mo. Median gain is under a month; the honest number is 12-mo OS 25.9% vs 11.4%
SWOG S1815	Adding nab-paclitaxel to gem/cis — negative . OS 14.0 vs 13.6 mo. Responses rose, survival did not
BILCAP	Adjuvant capecitabine — ITT p=0.097, primary endpoint not met ; standard of care on secondary analyses
ASCOT	Adjuvant S-1 , Japan — 3-yr OS 77.1% vs 67.6%. Not applicable in the US
FIGHT-202 / FOENIX-CCA2	Pemigatinib (ORR 37%) and futibatinib (ORR 42%) in FGFR2-altered disease
ReFocus	Lirafugratinib — ORR 46%, DOR 11.8 mo, far less hyperphosphatemia. Approved 23 Sep 2026
ClarIDHy	Ivosidenib in IDH1-mutant — PFS 2.7 vs 1.4 mo, HR 0.37
HERIZON-BTC-01	Zanidatamab in HER2 IHC 3+ — ORR 52%, DOR 14.9 mo
ROAR	Dabrafenib + trametinib in BRAF V600E — ORR 47–53%